

Ecosystem

1. Greater biological diversity of tropics than temperate region is due to the :

- (1) Presence of more seasonal environment
- (2) Frequent glaciations in the past
- (3) Highly variable climate and availability of less solar energy in the past
- (4) Availability of more solar energy which contributes to higher productivity

2. Read the following Assertion (A) and Reason (R) statements

Assertion (A): In Ecosystem Net primary productivity is less than gross primary productivity

Reasons (R) : Gross primary productivity minus photosynthesis losses is the net primary productivity

Choose the correct option from the options given below

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

3. Which one of the following is the correct statement

- (1) A given species may occupy more than one trophic level in the same ecosystem
- (2) A given species always occupy only one trophic level in the same ecosystem
- (3) A given species may not occupy more than one trophic level in the same ecosystem
- (4) Both (1) and (2)

4. Which type of ecological pyramid would be obtained with the following data

Secondary consumer- 2500 Joule

Primary consumer- 4500 Joule

Primary producer- 5000 Joule

Tertiary consumer- 1000 Joule

Choose the correct option

- (1) Upright pyramid of energy
- (2) Inverted pyramid of energy
- (3) Upright pyramid of number
- (4) This pyramid is not possible.

5. The annual net primary productivity (dry weight) of the whole biosphere is approximately :

- (1) 150 million tons
- (2) 150 billion tons
- (3) 170 million tons
- (4) 170 billion tons.

6. Study the food chain given in the box:

Dead Organic matter → Earthworm → Sparrow → Falcon

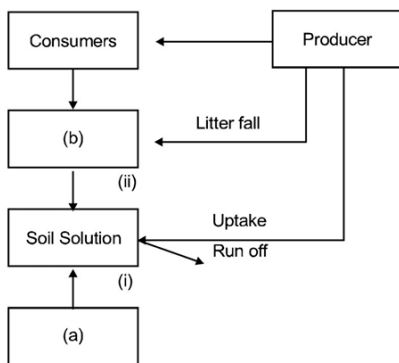
What type of food chain it is ?

- (1) Parasite
- (2) Detritus
- (3) Grazing
- (4) None

7. If we study productivity which of the following statement is incorrect :

- (1) Gross primary productivity of an ecosystem is the rate of production of organic matter by producers during photosynthesis
- (2) The annual net primary productivity of the whole biosphere is approximately 170 billion tons dry weight of organic matter
- (3) Net primary productivity is the available biomass for the consumption to producer
- (4) Secondary productivity is the rate of formation of new organic matter by consumers

8. In below simplified model of phosphorus cycling in a terrestrial ecosystem (a), (b), (i), (ii) are respectively



- (1) (a) Rock minerals, (b) Detritus (i) Weathering (ii) – Decomposition
 (2) (a) Detritus, (b) Rock minerals (i) Weathering (ii) – Decomposition
 (3) (a) Rock minerals, (b) Detritus (i) Decomposition (ii) – Weathering
 (4) (a) Detritus (b) Rock minerals (i) Decomposition (ii) – Weathering

9. Which statements is/are correct :

- (A) In most ecosystem all the pyramids of number, of energy and biomass are upright
 (B) Energy at a lower trophic level is always more than at a higher level
 (C) The pyramid of biomass in sea is generally inverted because the biomass of fishes far exceed that of phytoplankton
 (D) Saprophytes are not given any place in ecological pyramids even though they play vital role in the ecosystem

- (1) Only A & C
 (2) Only C & D
 (3) Only A, B & C
 (4) A, B, C & D

10. Ecosystem has two components namely

- (1) Plants and animals
 (2) Weeds and trees
 (3) Biotic and abiotic
 (4) Man and animal

11. The _____ of an ecosystem is the rate of production of organic matter during photosynthesis

- (1) Net primary productivity
 (2) Secondary production
 (3) Gross primary productivity
 (4) None of these

12. Consider the following statements and choose the correct option :-

- (i) Sun is the only source of energy for all ecosystem on earth
 (ii) PAR is less than 50 percent of incident solar radiation

- (1) only i is correct
 (2) only ii is correct
 (3) both i & ii are correct
 (4) both are incorrect

13. Build up of organic material in soil is under which of the following conditions to

- (1) < 10°C temperature, Absence of oxygen
 (2) Warm temperature, Humid environment
 (3) Aerobic conditions, > 25°C temperature
 (4) Nitrogen rich detritus, Optimum moisture

14. Humification results in the formation of _____ substance.

- (1) Colloidal, Basic
 (2) Acidic, Dark coloured
 (3) Amorphous, Light coloured
 (4) Light coloured, Colloidal

15. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The annual net primary productivity of the whole biosphere is approximately 170 billion tons (dry weight) of organic matter.

Reason (R) : Majority of this is contributed by the oceans as they occupy larger area of earth.

In the light of the above statement, choose the correct answer from the options given below:

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

16. _____ favours decompositions

- (1) Warm & dry environment
- (2) Warm & moist environment
- (3) cold & dry environment
- (4) cold & moist environment

17. Mineralisation is performed by

- (1) Small carnivores
- (2) Detrivores
- (3) Saprophytic bacteria and fungi
- (4) Earthworm, termites

18. Humus is :-

- (i) Dark coloured amorphous substance
 - (ii) Highly resistant to microbial action and undergoes decomposition at an extremely slow rate
 - (iii) Being colloidal in nature
 - (iv) Acts as a reservoir of nutrients
- (1) i, ii are incorrect
 - (2) i, ii & iv are correct but iii incorrect
 - (3) i, ii, iii, iv are correct
 - (4) All are incorrect

19. Read the following statements and select the correct option.

Statement A : Gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis.

Statement B : Net primary productivity is the available biomass for the consumption to herbivores and decomposers.

Select the correct option.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both the statements A and B are correct
- (4) Both the statements A and B are incorrect

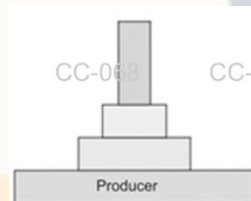
20. The annual net primary productivity of the whole biosphere is approximately

- (1) 190 million tons
- (2) 170 million tons
- (3) 170 billion tons
- (4) None of these

21. Major conduit of energy in land and aquatic ecosystem is transferred through _____ and _____ respectively.

- (1) GFC, DFC
- (2) PFC, GFC
- (3) DFC, GFC
- (4) GFC, GFC

22.



Which of the following does not hold true for the given ecological pyramid?

- a. Represents flow of energy in a pond ecosystem.
- b. Represents the amount of biomass at each trophic level in a lake ecosystem when producers are phytoplanktons.
- c. Represents the number of individuals at each trophic level in a forest ecosystem.
- d. Cannot represent number of individuals at each trophic level in a tree ecosystem.
- e. Cannot represent the flow of energy in a grassland ecosystem.

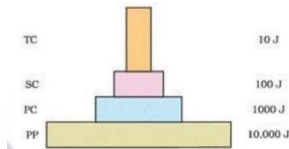
Choose the correct option.

- (1) b and d
- (2) c, d and e
- (3) a, c and d
- (4) b and e

23. Pyramid of energy is

- (1) Always inverted
- (2) Sometime upright
- (3) Always upright
- (4) Sometimes inverted

24. Identify the pyramid



- (1) Pyramid of number
- (2) Pyramid of biomass
- (3) Pyramid of energy
- (4) None of these

25. Match list-I with list-II

List-I	List-II
(a) Producer	(i) Wolf
(b) Primary consumer	(ii) Lion
(c) Secondary consumer	(iii) Grass
(d) Tertiary consumer	(iv) Grass hopper

Choose the correct answer from the options given below :

- (1) a-iii, b-ii, c-i, d-iv
- (2) a-iii, b-iv, c-i, d-ii
- (3) a-i, b-iv, c-ii, d-iii
- (4) a-ii, b-iii, c-i, d-iv

26. In ecosystem, GFC stands for

- (1) Generic flow control
- (2) Global fund for children
- (3) Grazing food chain
- (4) None of these

27. If 10,000 J of energy fall on piece of land, then what is energy available at T2 level?

- (1) 1000 J
- (2) 10 J
- (3) 100 J
- (4) 1 J

28. Choose incorrect option w.r.t. amount of living material present in different trophic levels at a given time

- (1) Can be measured as number
- (2) Is equivalent to standing crop
- (3) Is always represented as dry weight only
- (4) Expressed both as biomass and number

29. Given below are two statements

Assertion (A) : Ecosystems are exempted from the second law of thermodynamics.

Reason (R) : Ecosystems need constant supply of energy to synthesize the molecule they require to counteract the universal tendency towards increasing disorderliness.

Choose the most appropriate answer

- (1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is wrong.
- (3) (A) is incorrect but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

30. All of the following are true for humus, except

- (1) Rich in cellulose, lignin and tannins
- (2) Colloidal in nature
- (3) Susceptible to microbial action
- (4) Slightly acidic in nature

31. Rate of production of organic matter by producers per unit time and area is

- (1) GPP
- (2) NPP
- (3) $NPP - R$
- (4) More than one option is correct

32. Read the following statement a to d.

(a) An ecosystem can be visualised as a functional unit of nature.

(b) Ecosystem varies greatly in size from a small pond to a large forest or a sea.

(c) Many ecologists regard the entire biosphere as a global ecosystem

(d) Crop field & an aquarium may also be considered as natural ecosystem

How many of the above statements are correct.

- (1) 1
- (2) 2
- (3) 3
- (4) 4

33. Vertical distribution of different species occupying different levels is called as

- (1) Standing crop
- (2) Standing state
- (3) Stratification
- (4) Decomposition

34. What percentage of PAR is captured by plants?

- (1) More than 50%
- (2) 50%
- (3) 20-40%
- (4) 2-10%

35. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R) .

Assertion (A) : The available biomass for the consumption to heterotrophs is known as secondary productivity.

Reason (R) : A considerable amount of GPP is utilised by plants in respiration.

Choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is wrong.
- (3) (A) is incorrect but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

36. The correct way of decomposition

- (1) Fragmentation → leaching → humification → catabolism → mineralization
- (2) Fragmentation → leaching → catabolism → humification → mineralization
- (3) Fragmentation → catabolism → leaching → mineralization → humification
- (4) Fragmentation → mineralization → catabolism → leaching → humification

37. The rate of biomass production is called productivity. It is expressed in terms of ____.

- (1) g m^{-2}
- (2) Kcal m^{-2}
- (3) $\text{g m}^{-2} \text{ yr}^{-1}$
- (4) Kcal yr^{-1}

38. Which is not true regarding ecosystem?

- (1) Self sufficient unit
- (2) Cyclic exchange of materials between living beings and environment
- (3) The only requirement is input of the energy
- (4) Characterized by a major vegetation type

39. Which of the following statement is not true for humus?

- (1) Dark coloured amorphous substance
- (2) Highly resistant to microbial action
- (3) Act as reservoir of nutrients and increases water holding capacity of soil
- (4) They are degradation product of protein and fats and are produced by the process of mineralisation

40. Which of the following phenomena does not play a significant role in the maintenance of homeostasis in an ecosystem?

- (1) Carrying capacity
- (2) Biotic potential
- (3) Self regulation
- (4) Feed back

41. Which of the following trophic levels are occupied by sparrow?

- (1) Primary consumer, primary producer
- (2) Secondary consumer, top carnivore
- (3) Primary producer, secondary producer
- (4) Primary consumer, secondary consumer

42. Given below are two statements:

Statement-I : Pyramid of energy is always upright.

Statement-II : A given species may occupy more than one trophic level in the same ecosystem at the same time.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement-I and II are incorrect.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

43. The open ocean and tropical rain forest are the largest contributors to Earth's net primary productivity because

- (1) Both have high rates of net primary productivity
- (2) Both cover huge surface areas of the Earth
- (3) Nutrient cycle occur fastest in these two ecosystem
- (4) The ocean covers a huge surface area and the tropical rain forest has a high rate of productivity

44. If 20J of energy is trapped at producer level, then how much energy will be available to peacock as food in following chain?

Plant → Mice → Snake → Peacock

- (1) 0.02 J
- (2) 0.2 J
- (3) 0.002 J
- (4) 0.0002 J

45. Identify the correct set of statements :

- (a) Mass of living material at each trophic level in a particular time is called standing crop.**
- (b) Number of trophic level in the grazing food chain is restricted.**
- (c) In an aquatic ecosystem, GFC is the major conduit for energy flow.**
- (d) Inter connected food chains make food web.**
- (e) Detritus food chain begins with primary producers.**

Choose the correct answer

- (1) a, b and c only
- (2) a, c and d only
- (3) a, b and e only
- (4) a, b, c and d

46. Which of the following food chain shows spindle shaped pyramid of number?

- (1) Grass - Insects - Frog
- (2) Phytoplankton - Zooplankton - Small fish
- (3) Tree - Birds - Hawks
- (4) Tree - Birds - Insects - Bacteria

47. An ecosystem refers to

- (1) A group of species interacting with each other in terrestrial environment but not in aquatic environment
- (2) An area comprising only physical factor temperature, soil, etc. like
- (3) The group of living organisms interacting with each other and their physical environment
- (4) Interaction of only biotic components among themselves in a particular habitat

48. In an ecosystem, 100 kJ energy in the form of sunlight has fallen on the organism that occupies the first trophic level of a food chain. What amount of that energy will be utilised by the organism for its different biological activities which occupies the second trophic level?

- (1) 0.009 kJ
- (2) 0.9 kJ
- (3) 9 kJ
- (4) 0.09 kJ

49. PAR stands for

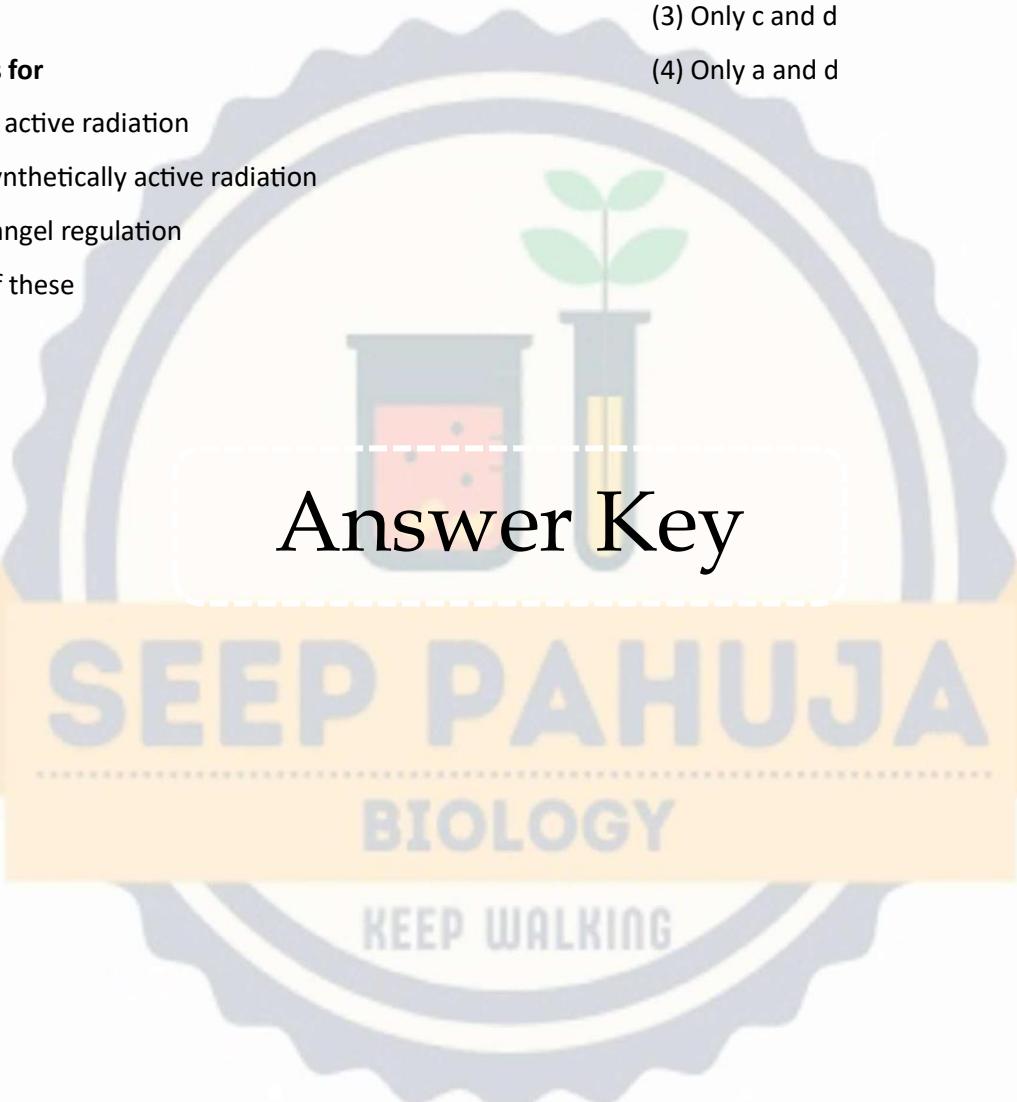
- (1) Percent active radiation
- (2) Photosynthetically active radiation
- (3) Power angel regulation
- (4) None of these

50. Which amongst the following are the characteristics of man made ecosystems?

- a. Abundant cycling of nutrients
- b. Simple food chain
- c. Little diversity
- d. Low productivity

The correct ones are

- (1) Only a and b
- (2) Only b and c
- (3) Only c and d
- (4) Only a and d



Answer Key

- 1. (4)
- 2. (2)
- 3. (1)
- 4. (1)
- 5. (4)
- 6. (2)
- 7. (3)
- 8. (1)
- 9. (4)
- 10. (3)
- 11. (3)
- 12. (2)
- 13. (1)
- 14. (2)
- 15. (2)
- 16. (2)
- 17. (3)
- 18. (3)
- 19. (3)
- 20. (3)
- 21. (3)

SEEP PAHUJA
BIOLOGY
KEEP WALKING

- 22. (4)
- 23. (3)
- 24. (3)
- 25. (2)
- 26. (3)
- 27. (1)
- 28. (3)
- 29. (3)
- 30. (3)
- 31. (1)
- 32. (3)
- 33. (3)
- 34. (4)
- 35. (3)
- 36. (2)
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- 39. (4)
- 40. (2)
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- 42. (4)
- 43. (4)
- 44. (1)
- 45. (4)
- 46. (3)
- 47. (3)
- 48. (4)
- 49. (2)
- 50. (2)

